DAY 16 | PRE-READ

Asking Data Good Questions

A light read before Day 16. About 15 minutes, no code required.

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Why read this?

Tomorrow you bring together everything from the data days into one pipeline that could run every night at a real company. You will also refresh SQL, the most widely used language for asking questions of data.

1. SQL is a way of asking questions

SQL reads almost like English. "Show me the city and total sales, from the orders table, only for paid orders, grouped by city, biggest first." That sentence is nearly a SQL query already. Analysts, engineers and many business tools all speak SQL.

Plain English	SQL word
Show me these columns	SELECT
From this table	FROM
Only rows where	WHERE
Grouped by	GROUP BY
Only groups where	HAVING
Sorted by	ORDER BY

2. From a one-off script to a daily job

A recipe you cook once for yourself can be messy. A recipe a restaurant cooks every night must be written down, use measured ingredients, and include checks ("taste before serving"). A production data pipeline is the restaurant version: it takes the date as an input, checks its own results, and keeps a record of every run.

Key idea: a good pipeline can be re-run for any day and always gives the same correct answer.

3. Checks before serving

- Is there any data at all today?
- Are there duplicate orders?
- Do the totals in the summary match the detailed data?

If a check fails, the pipeline should stop loudly rather than quietly publish wrong numbers. Wrong numbers that look right are the most dangerous kind.

4. Looking back over three weeks

You have learned Python, built APIs, stored data in SQL and MongoDB, secured endpoints, built React screens and processed data with Spark. Tomorrow we connect these pieces into one picture before moving on to the cloud next week.

5. Think about these before class

- Write three questions your team would love to answer from its data, in plain English.
- What checks would you want before a sales report goes to your CEO?
- Which topic from the last three weeks would you most like to revisit?

Furikaeri (振り返り): looking back to reflect and improve